A finite degree-elevation bound for central Bernstein polynomials

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Abstract

Let n be a positive even integer and let $f(x) = k\binom{n}{n/2}x^{n/2}(1-x)^{n/2}$, where $k > 1$ and $h = f(1/2) < 1$. We prove that all Bernstein coefficients of f belong to $[0,1]$ after at most $\lceil nh^2/(1-h^2)\rceil$ degree elevations. This establishes the finite upper bound asked for by Peter Occil. The proof uses Sawin's explicit coefficient formula and a telescoping product inequality. A parity refinement gives $r_{\min} \le 2\lceil nh^2/(2(1-h^2))\rceil - 1$.

1 The question and the result

For an integer $N \ge 0$, the degree-N Bernstein basis is

$$B_{i,N}(x) = \binom{N}{i}x^i(1-x)^{N-i}, \qquad 0 \le i \le N.$$

Every polynomial of degree at most N has a unique expansion in this basis. Increasing N while keeping the polynomial fixed is called *degree elevation*.

Fix a positive even integer $n = 2m$ and a real number $k > 1$, and suppose

$$h = \frac{k\binom{2m}{m}}{4^m} \in (0,1).$$

The polynomial under consideration is

$$f(x) = k\binom{2m}{m}x^m(1-x)^m = 4^m h x^m(1-x)^m.$$

Its maximum on $[0,1]$ is h, attained at $1/2$. For an integer $r \ge 0$, write

$$f(x) = \sum_{i=0}^{2m+r} b_i^{(r)} B_{i,2m+r}(x),$$

and let $r_{\min}$ be the least r for which all these coefficients belong to $[0,1]$. At $r = 0$ the only nonzero coefficient is k, so $r_{\min} \ge 1$ whenever it exists.

Occil asked whether $\lfloor nh^2/(1-h^2)\rfloor + 1$ elevations always suffice [1]. Sawin derived the elevated coefficients and located their maximum; a central-binomial asymptotic then gives the scale $nh^2/(1-h^2)$ [2]. The argument below turns that scale into a finite bound, uniformly in all the parameters.

Theorem 1. *Under the hypotheses above, put $T = nh^2/(1-h^2)$. Then*

$$1 \le r_{\min} \le 2\left\lceil \frac{T}{2} \right\rceil - 1 \le \lceil T \rceil \le \lfloor T \rfloor + 1.$$

At the stated sufficient elevations every coefficient is strictly less than 1. Moreover, $r_{\min}$ is odd.

2 The maximum coefficient and its bound

We give the coefficient calculation in full, including the parity needed for the finite bound.

Lemma 2 (Coefficient formula and maximum)**.** *For $r \geq 0$, the nonzero Bernstein coefficients are*

$$b_{m+j}^{(r)} = 4^m h \frac{\binom{r}{j}}{\binom{2m+r}{m+j}}, \qquad 0 \leq j \leq r. \tag{1}$$

Their maximum is attained at $j = r/2$ if r is even, and at $j = (r-1)/2, (r+1)/2$ if r is odd. For $r \geq 1$, set $R = 2\lceil r/2 \rceil$. Then

$$\max_i b_i^{(r)} = h \prod_{\ell=1}^{m} \frac{R + 2\ell}{R + 2\ell - 1}. \tag{2}$$

Proof. Multiplication by $(x + (1-x))^r = 1$ gives

$$f(x) = 4^m h \sum_{j=0}^{r} \binom{r}{j} x^{m+j} (1-x)^{m+r-j},$$

which proves (1); all other coefficients are zero. For $0 \leq j < r$, the ratio of consecutive nonzero coefficients is

$$\frac{b_{m+j+1}^{(r)}}{b_{m+j}^{(r)}} = \frac{(r-j)(m+j+1)}{(j+1)(m+r-j)}.$$

The numerator minus the denominator on the right is $m(r-2j-1)$. This establishes the asserted maxima.

If $r = 2s$ is even, its maximum divided by h is

$$4^m \frac{\binom{2s}{s}}{\binom{2s+2m}{s+m}} = \prod_{\ell=1}^{m} \frac{2s + 2\ell}{2s + 2\ell - 1},$$

by cancellation of factorials. If $r = 2s - 1$ is odd, with $s \geq 1$, use

$$\binom{2s-1}{s-1} = \frac{1}{2}\binom{2s}{s}, \qquad \binom{2s+2m-1}{s+m-1} = \frac{1}{2}\binom{2s+2m}{s+m}.$$

Thus the maximum at elevation $2s-1$ equals the maximum at elevation $2s$, proving (2) in both cases. □

Lemma 3 (A telescoping bound)**.** *For every integer $r \geq 1$ and $R = 2\lceil r/2 \rceil$,*

$$\left(\max_i b_i^{(r)}\right)^2 < h^2 \frac{R+n}{R} \leq h^2 \frac{r+n}{r}. \tag{3}$$

Proof. For every real $a > 2$,

$$\left(\frac{a}{a-1}\right)^2 < \frac{a}{a-2},$$

since $(a-1)^2 = a(a-2) + 1$. Here $R \geq 2$, so this inequality applies to every $a = R + 2\ell$ in (2). Consequently

$$\prod_{\ell=1}^{m} \left(\frac{R+2\ell}{R+2\ell-1}\right)^2 < \prod_{\ell=1}^{m} \frac{R+2\ell}{R+2\ell-2} = \frac{R+2m}{R}.$$

Multiplication by h^2 proves the strict inequality in (3). The remaining inequality uses $R \geq r > 0$. □

Proof of Theorem 1. Let $q = \lceil T/2 \rceil$, so $q \geq 1$, and take $r = 2q - 1$. Then $R = 2q \geq T$, whence

$$h^2 \frac{R+n}{R} \leq 1 \quad \Longleftrightarrow \quad R \geq \frac{nh^2}{1-h^2}.$$

Lemma 3 now gives $\max_i b_i^{(r)} < 1$. Every coefficient is nonnegative by (1). This proves existence of $r_{\min}$ and the first upper bound, including the case $R = T$ because the product inequality is strict.

To compare the integer bounds, set $L = \lceil T \rceil$. If $L = 2a$, then $2\lceil T/2 \rceil - 1 \leq 2a - 1 \leq L$. If $L = 2a + 1$, then $2\lceil T/2 \rceil - 1 \leq 2a + 1 = L$. Finally $L \leq \lfloor T \rfloor + 1$. Each of these two larger elevations satisfies $r \geq T$, so (3) gives strict coefficient bounds at those elevations as well.

The case $r = 0$ fails because $k > 1$. Every positive even elevation $2s$ has exactly the same maximum as elevation $2s - 1$, by Lemma 2. Since all coefficients are nonnegative, the first successful elevation must therefore be odd. □

3 An exact criterion and examples

The product formula also describes the answer exactly, even when the simple sufficient bound is larger than the minimum.

Corollary 4. *Let s_* be the least positive integer s satisfying*

$$h \prod_{\ell=1}^{m} \frac{2s + 2\ell}{2s + 2\ell - 1} \leq 1.$$

Then $r_{\min} = 2s_* - 1$.

Proof. Existence follows from Theorem 1. The displayed product times h is the common maximum at elevations $2s - 1$ and $2s$. All coefficients are nonnegative, and elevation zero fails. The criterion and the formula for $r_{\min}$ follow. □

For example, take $n = 2$ and $h = 3/4$. Then $k = 3/2$ and $T = 18/7$. At elevation $r = 1$, the four coefficients are $(0, 1, 1, 0)$, so $r_{\min} = 1$. The requested bound gives $r_{\min} \leq 3$. This example also shows that the asymptotic scale T need not be a lower bound for the exact minimum.

For an integer-boundary example, take $n = 32$ and $h = 3/5$. These parameters satisfy $k > 1$: indeed $\binom{32}{16}/2^{32} = 601080390/4294967296 < 3/5$. Here $T = 18$, and the parity bound gives $r_{\min} \leq 17$. The strict inequality in Lemma 3 applies with $R = T = 18$, so this boundary case requires no limiting argument.

The proof is entirely algebraic. The accompanying exact-arithmetic program checks the coefficient formula against repeated degree elevation, the maximizing indices, the product identity, and the stated bounds on finite sets of parameters. These checks provide reproducible verification of the formulas used in the proof.

References

[1] Peter Occil, *On the degree elevation needed to bring Bernstein coefficients to* $[0, 1]$, MathOverflow question 381419, 17 January 2021, revised the same day. mathoverflow.net/q/381419.

[2] Will Sawin, Answer to *On the degree elevation needed to bring Bernstein coefficients to* $[0, 1]$, MathOverflow, 17 January 2021, edited 18 January 2021. mathoverflow.net/a/381465.